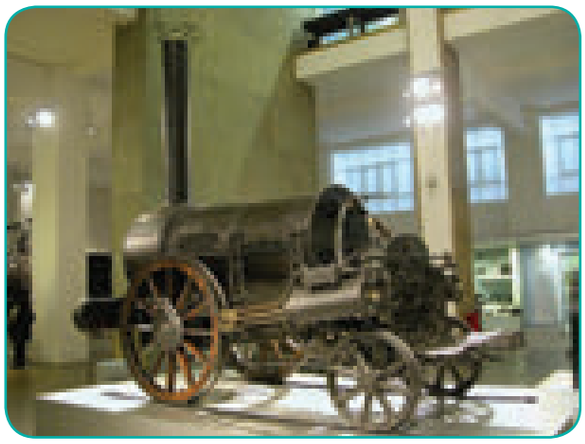
Steam Engines: The steamboat preceded the steam engine as a means of locomotion. On the Firth of Clyde Canal there was a steam boat in 1802. In 1804 the first locomotive was made. In 1830 the first passenger railway between Liverpool and Manchester was opened. George Stephenson's engine, "The Rocket," functioned with a speed of over thirty miles an hour, unimaginable at that time.



Steam Engine - The Rocket

In 1807, an American Robert Fulton made the first successful steam boat. In April 1838, the first steasmships, the *Sirius* and the *Great Western*, crossed the Atlantic. Isambard Kingdom Brunel, an English Engineer, built the first fully iron ship with the screw propellers called SS *Great Britain* in 1843. In earlier times, instead of screw propellers, paddle wheels were used.

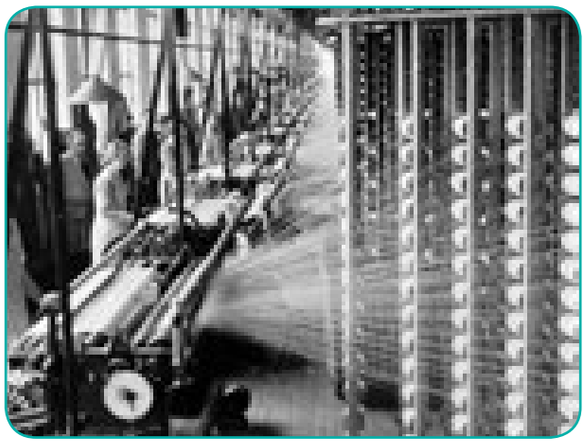
Roads: With the increase in production, it became important to have good roads. However, the roads were of poor quality and the travelling time was long and strenuous. Due to the pressure exercised by leading industrialists roads were maintained by turnpikes, who collected toll from the people for the proper management of the roads. John Loudon McAdam invented an effective and economical method of constructing roads.

Macadamisation means to pave the road by laying and compacting successive layers of broken stone, often with hot tar.



John McAdam

In 1835 the first electric telegraph came into existence. Sixteen years later the first undersea cable was laid between England and France. In a few years the telegraph system spread throughout the world. The modern factory with its giant chimneys began to dominate the landscape of the area around Manchester in Lancashire and Glasgow in Scotland. In 1750, England had two cities with more than 50,000 inhabitants, London and Edinburgh. By 1851 the number of cities of this size had increased to 29.



Industrial Manchester

Second Stage of Industrial Revolution : Germany and the USA

The significant discoveries of the Second Industrial Revolution emanated more from the laboratory of the physicist or chemist than from the brain of the individual inventor. The other essential features of the Second Industrial Revolution were the introduction of automatic machinery, and the enormous increase in mass production and a division of the labour into minute segments of the manufacturing process.

Throughout the eighteenth century there was steady industrial development and great commercial activity in Western Europe. This was exemplified by the development of banking, and improvement in internal

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